

Machine Learning Strategy for Coupons' Offer

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Business Context

Problem

Increase customer engagement by offering the most effective coupon for each customer.

Available data

Offers: Information about each type of coupon;

Profile: Customer's information;

Transactions: Customer purchase events, including interactions related to received offers.

Proposed solution

Predict whether a customer will complete an offer (i.e., make a purchase using a coupon) within its validity period, and identify which factors most influence conversion.

ML Strategy Proposal

ML Solution

Binary classification model that predicts whether a customer will convert or not if he/she receives an offer within its valid period.

Model insights

- The model is able to identify converting customers **93% of the time** (recall).
- ***The model maximizes overall conversions:***
 - *Advantage:* we almost never miss customers who would convert;
 - *Disadvantage:* we will likely send offers to customers who will not convert, which may generate additional costs;
- We can discuss how conservative we want to be when sending coupons and adjust the model accordingly to match that strategy.

What are the key factors that contributes to conversion?

- **Duration:** Offers with longer validity period are more likely to be completed;
- **Customer lifetime value:** Customers who already spend more are more likely to accept coupons;
- **Discount value:** There are different discount values that lead to conversion (not necessarily the higher the discount the better);
- **Offer type;**
- **Credit card limit:** The higher the credit card limit, it is more likely the customer will accept the offer.

Recommendation Summary

A/B testing for offer duration

Explore longer offer durations for the different types of coupons to validate whether this update may increase overall conversion.

Future Project Ideias

Customer clustering

Identify customers with similar behavior to understand which groups are more impacted by specific offer characteristics.

E.g.: heavy vs. low buyers, different income ranges (potentially related to their credit card limits).

Rank best offer type

Given a list of offer types, calculate the conversion probability for customer A for each offer X, Y, Z — and send the offer with the highest conversion probability.